

Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

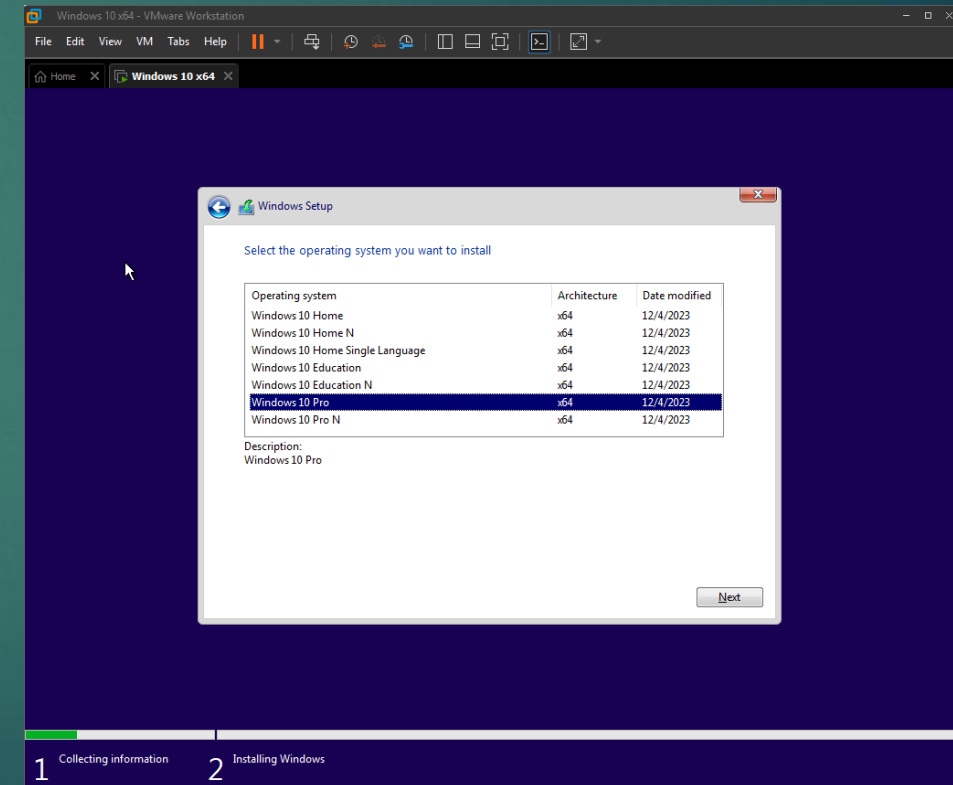
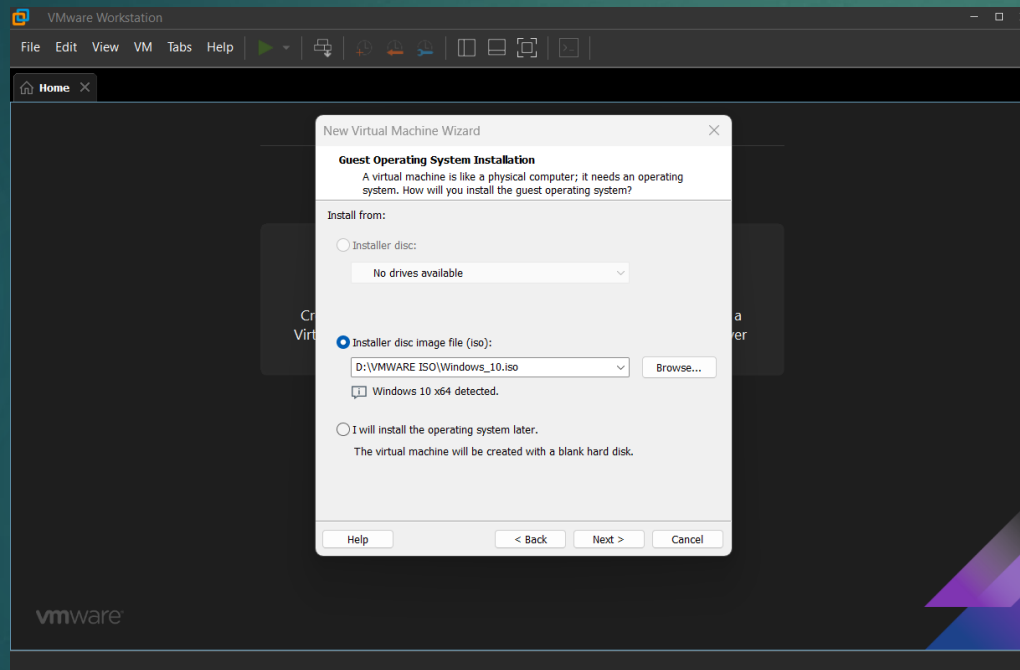
Objectives:

- Setup Windows 10 and Ubuntu VMs using a hypervisor (e.g., Vmware Workstation or Hyper-V)

Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

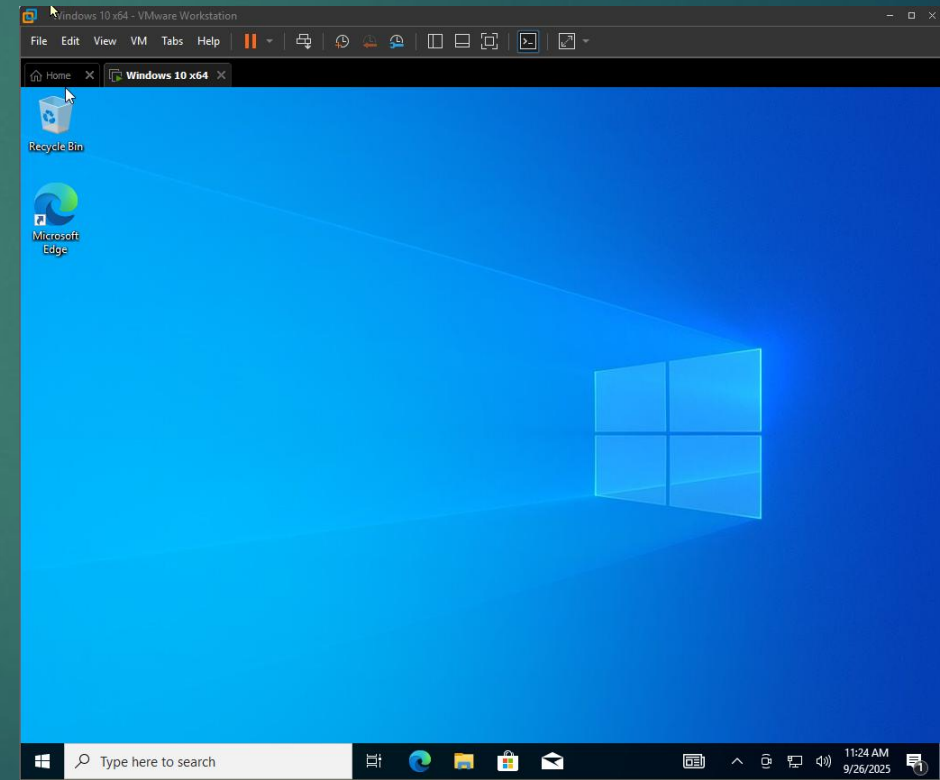
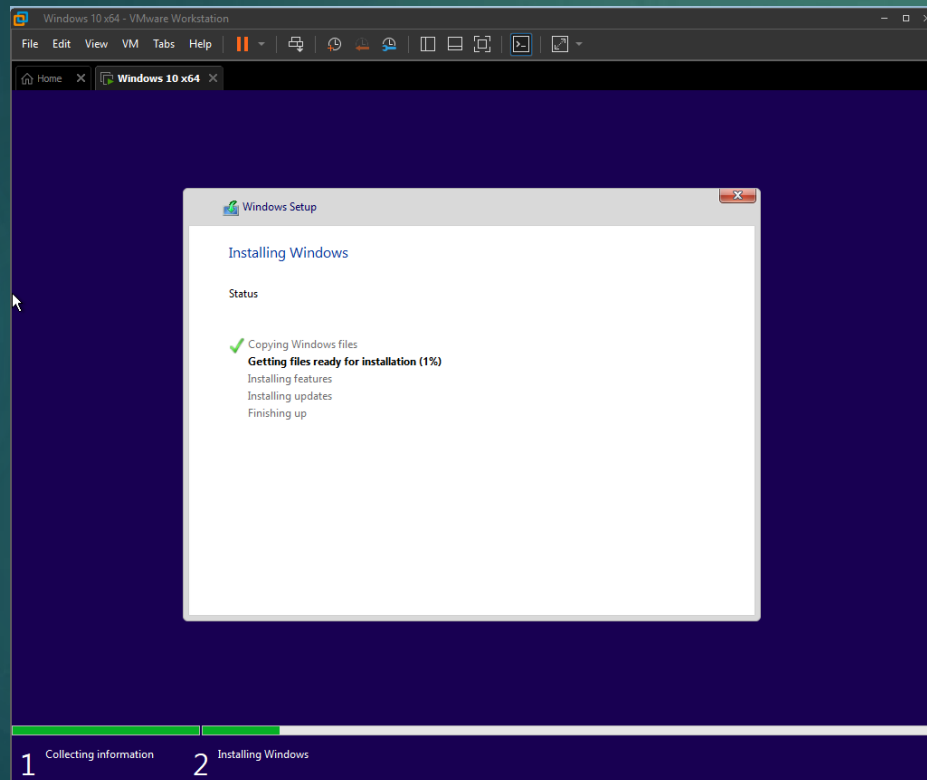
Installing Windows 10 in VMware Workstation



Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

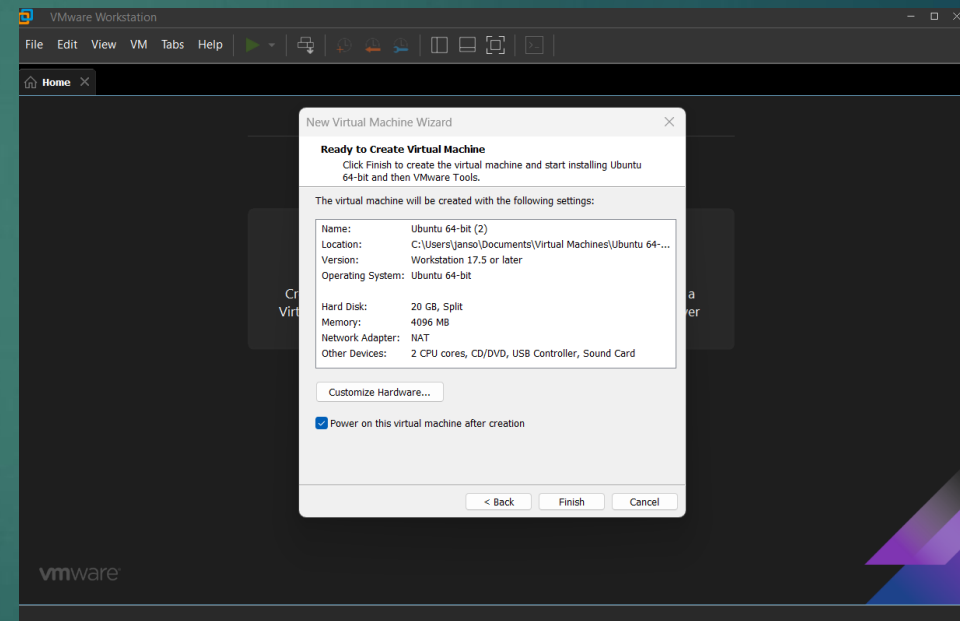
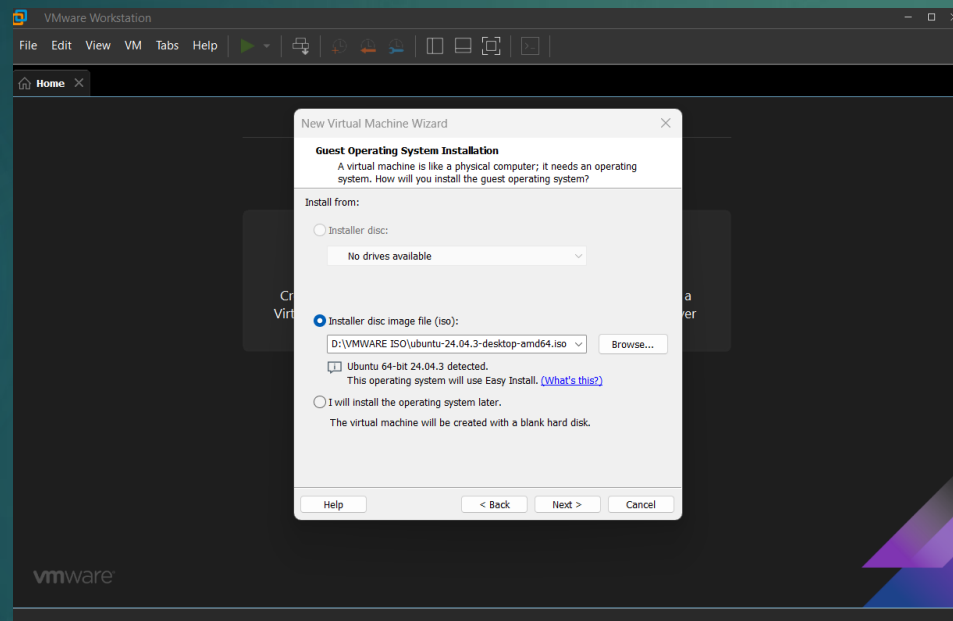
Installing Windows 10 in Vmware Workstation - continue



Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

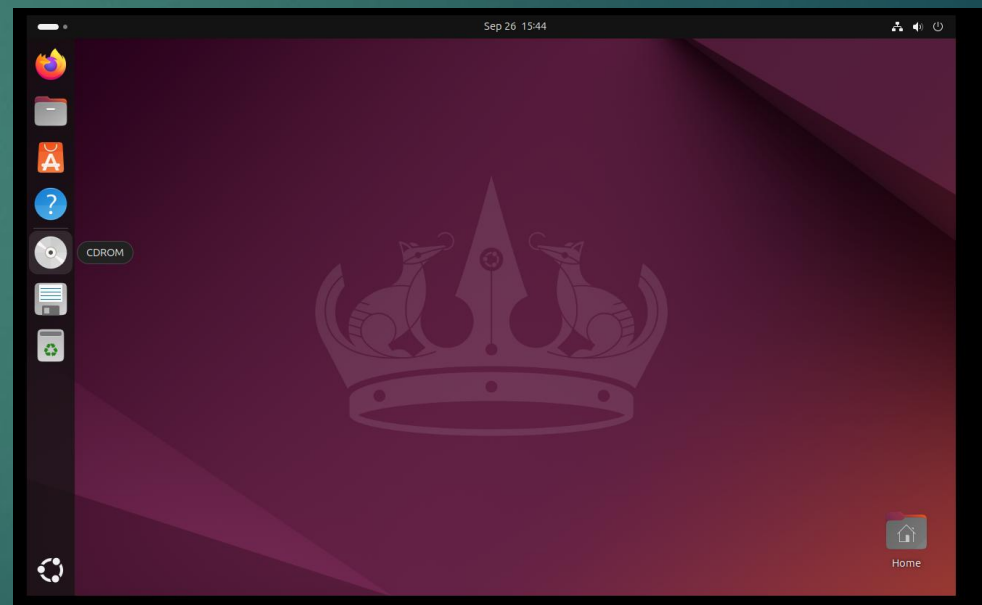
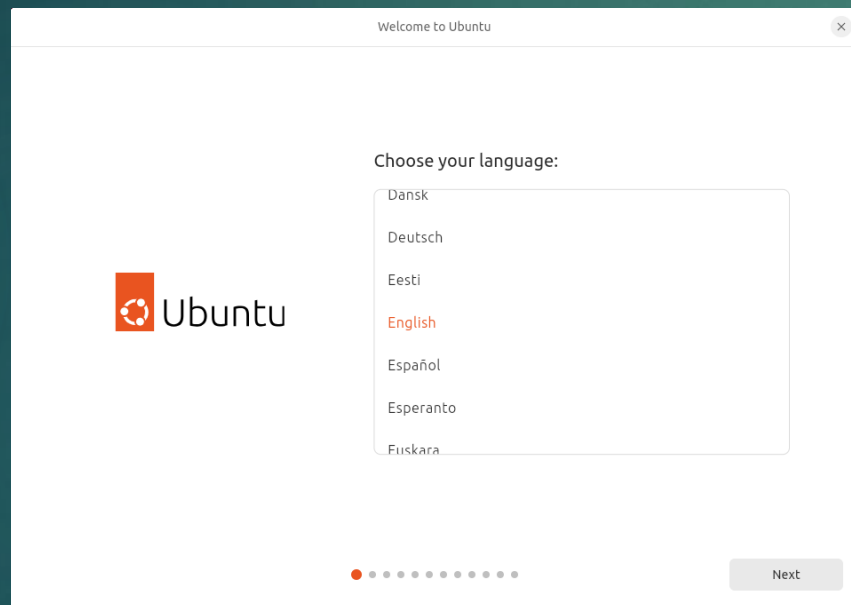
Installing Ubuntu in Vmware Workstation



Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

Installing Ubuntu in Vmware Workstation - continue



Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

Ping Result of Ubuntu and Windows 10

Ubuntu IP address (192.168.121.130)

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:70:05:8b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.121.130/24 brd 192.168.121.255 scope global dynamic noprefixroute ens33
        valid_lft 1650sec preferred_lft 1650sec
    inet6 fe80::20c:29ff:fe70:58b/64 scope link
        valid_lft forever preferred_lft forever
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Windows IP Address (192.168.121.129)

```
Ethernet adapter Ethernet0:

Connection-specific DNS Suffix  . : localdomain
Description . . . . . : Intel(R) 82574L Gigabit Network Connection
Physical Address. . . . . : 00-0C-29-89-D3-35
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::d01b:84c6:bc0e:5de1%12(Preferred)
IPv4 Address. . . . . : 192.168.121.129(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Tuesday, 30 September, 2025 04:00:28 PM
Lease Expires . . . . . : Tuesday, 30 September, 2025 04:30:28 PM
Default Gateway . . . . . : 192.168.121.2
DHCP Server . . . . . : 192.168.121.254
DHCPv6 IAD . . . . . : 100666409
DHCPv6 Client DUID. . . . . : 00-01-00-01-30-68-8B-C2-00-0C-29-89-D3-35
DNS Servers . . . . . : 192.168.121.2
Primary WINS Server . . . . . : 192.168.121.2
NetBIOS over Tcpip. . . . . : Enabled
PS C:\Windows\system32>
```

Ubuntu Ping result

(Ping to Window 10, 192.168.121.129)

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ ping 192.168.121.129
PING 192.168.121.129 (192.168.121.129) 56(84) bytes of data:
64 bytes from 192.168.121.129: icmp_seq=1 ttl=128 time=0.641 ms
64 bytes from 192.168.121.129: icmp_seq=2 ttl=128 time=1.89 ms
64 bytes from 192.168.121.129: icmp_seq=3 ttl=128 time=1.61 ms
64 bytes from 192.168.121.129: icmp_seq=4 ttl=128 time=2.15 ms
64 bytes from 192.168.121.129: icmp_seq=5 ttl=128 time=1.57 ms

--- 192.168.121.129 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4041ms
rtt min/avg/max/mdev = 0.641/1.572/2.149/0.510 ms
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Windows Ping Result

(Ping to Ubuntu 192.168.121.130)

```
PS C:\Windows\system32> ping 192.168.121.130

Pinging 192.168.121.130 with 32 bytes of data:
Reply from 192.168.121.130: bytes=32 time<1ms TTL=64
Reply from 192.168.121.130: bytes=32 time=1ms TTL=64
Reply from 192.168.121.130: bytes=32 time=1ms TTL=64
Reply from 192.168.121.130: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.121.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
PS C:\Windows\system32>
```

Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

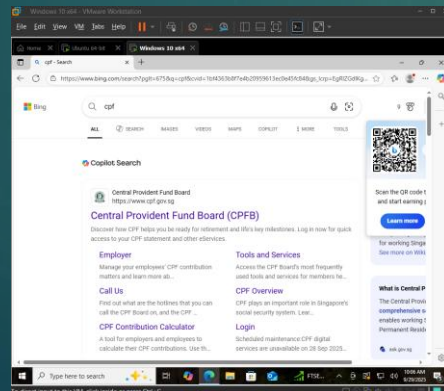
Local Machine Ping Result to Ubuntu and Windows 10

Local Machine IP Address (192.168.121.1)

Ethernet adapter VMware Network Adapter VMnet8:

```
Connection-specific DNS Suffix . : 
Description . . . . . : VMware Virtual Ethernet Adapter for VMnet8
Physical Address. . . . . : 00-50-56-C0-00-08
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::c12c:2acc:9f79:65d9%9(Preferred)
IPv4 Address. . . . . : 192.168.121.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 
DHCPv6 IAID . . . . . : 1174425686
DHCPv6 Client DUID. . . . . : 00-01-00-01-27-DE-CE-B1-08-97-98-E6-7D-26
NetBIOS over Tcpip. . . . . : Enabled
```

Using VM Windows to Surf Internet



Local Machine Ping Result to Ubuntu and Windows

```
PS C:\Users\janso> ping 192.168.121.130
```

```
Pinging 192.168.121.130 with 32 bytes of data:
Reply from 192.168.121.130: bytes=32 time<1ms TTL=64
Reply from 192.168.121.130: bytes=32 time<1ms TTL=64
Reply from 192.168.121.130: bytes=32 time<1ms TTL=64
Reply from 192.168.121.130: bytes=32 time<1ms TTL=64
```

```
Ping statistics for 192.168.121.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\janso> ping 192.168.121.129
```

```
Pinging 192.168.121.129 with 32 bytes of data:
Reply from 192.168.121.129: bytes=32 time=2ms TTL=128
Reply from 192.168.121.129: bytes=32 time=1ms TTL=128
Reply from 192.168.121.129: bytes=32 time<1ms TTL=128
Reply from 192.168.121.129: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.121.129:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 2ms, Average = 0ms
PS C:\Users\janso>
```


Project 2: Implementation Workstation Solution

Part 2: Deploying Virtual Machines

Troubleshooting Journey & Experience

VM was fun, but ran into few problems. First, I encounter my VM were really slow and lag, due to small amount of memory allocated. Once I increased those settings, they work much better.

Second, I was unable to ping from Ubuntu to Windows 10 because firewall is blocking. I went into Windows Defender setting, toggle it off and the ping was successful.

Overall, the most common problems were slow performance, network issues, and firewall settings. Once I figured those out, everything worked great.

Project 2: Implementation Workstation Solution

Part 3: The windows 11

Objectives:

- Research the benefits of Windows 11

Project 2: Implementation Workstation Solution

Part 3: The windows 11

Windows Timeline

Product name	Release date
Windows 1.0	November 20, 1985
Windows 2.0	December 9, 1987
Windows 2.1	May 27, 1988
Windows 3.0	May 22, 1990
Windows 3.1	April 6, 1992
Windows 3.2	November 22, 1993
Windows 95	August 24, 1995
Windows 98	June 25, 1998
Windows 98 SE	May 5, 1999
Windows 2000	February 17, 2000
Windows Me	September 14, 2000
Windows XP	October 25, 2001
Windows XP 64-bit Edition	March 28, 2003
Windows XP Professional x64 Edition	April 25, 2005
Windows Vista	January 30, 2007
Windows 7	October 22, 2009
Windows 8	October 26, 2012
Windows 8.1	October 17, 2013
Windows 10	July 29, 2015
Windows 11	October 5, 2021

Project 2: Implementation Workstation Solution

Part 3: The windows 11

Differences Between Windows for PC and Windows for Server

Features	Windows for PC	Windows for Server
Purpose	For personal use, general computing, entertainment, browsing internet, gaming, etc	For enterprise-level, hosting services, managing network, handling databases, file sharing, etc.
User Interface	Graphic User Interface (GUI), best use with touch and mouse.	Minimal GUI, command line interface, remote management.
Hardware Support	Consumer grade hardware, limited ram and cores. More to graphics performance and user experience.	Using high-end processors, large amount of RAM. Using hot-swappable drive.
Concurrent Network Connection	Limited to 10 – 20 connections	Support thousands of concurrent connections for enterprise use.
Licensing and Cost	Lower cost, consumer licensing	More costly, licensing model for businesses that require administrative control and support for multiple users.

Project 2: Implementation Workstation Solution

Part 3: The windows 11

Justify Environment

	Windows for PC	Windows for Server
Best For	Home user, students, office workers, developer, and general computing.	Enterprises, data centers, cloud infrastructure.
Environment	Personal desktop, laptop, small office workstation, school.	Backend server, domain controller, file server, web server, virtualization host.
Why	Provide rich user experiences, support wide range of consumer applications, and easy to use and maintain.	Reliability, scalability, centralize management of resources across network.

Project 2: Implementation Workstation Solution

Part 3: The windows 11

Different Operating System Comparison Table

	Windows	Linux	Mac OS	Others
Market Population(%)	69.75	3.93	13.48	12.84
Frequency of system updates or patches release	Second Tuesday of each month	Starting from REHL 8, minor release are plan for every six months.	Yearly, typically in October or September.	4 – 6 weeks
Which Operating System (OS) has the highest exploitaion (Use scale from 1 – 4)	4	2	2	1
Which Operating System has the highest vulnerabilities (use scale from 1 – 4)	4	3	2	1

Project 2: Implementation Workstation Solution

Part 3: The windows 11

Understanding OS Vulnerabilities

Operating System	Vulnerability
Windows	More prone to exploitation usually they are more popular, and usage is very high, and hackers focus on them due to more user
Linux	Less popular in desktop, but heavily exploited in enterprise and cloud environments. Open source and backend role, makes them a target for hacker.
MacOS	Smaller user bases, and harder to break in as Apple controls both the hardware and software which makes them more safer

Why Windows 11 should be used in a business environment

- Works well with Microsoft Office, Teams, and Clouds tools. Good for teamwork and remote work.
- Fast and smooth helps employees to stay productive.
- Easy for IT teams to manage and update across many computers.
- Enhance security, has built in tools to block virus and protect data.

Project 2: Implementation Workstation Solution

Part 4: Linux

Objectives:

- Understanding Linux CLI Command to Manage Linux Machine

Project 2: Implementation Workstation Solution

Part 3: Linux

Verify the Installed OS Version

OS Version: lsb_release -a

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Project 2: Implementation Workstation Solution

Part 3: Linux

Verify CPU Info

CPU Info: `lshw -c cpu`

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ lshw -c cpu
WARNING: you should run this program as super-user.
*-cpu:0
  product: Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz
  vendor: Intel Corp.
  physical id: 1
  bus info: cpu@0
  version: 6.165.2
  width: 64 bits
  capabilities: fpu fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mm
x fxsr sse sse2 ss syscall nx pdpe1gb rdtscp x86-64 constant_tsc arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cp
uid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervi
sor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap c
lflushopt xsaveopt xsavec xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
  configuration: microcode=4294967295
*-cpu:1
  product: Intel(R) Core(TM) i7-10750H CPU @ 2.60GHz
  vendor: Intel Corp.
  physical id: 2
  bus info: cpu@1
  version: 6.165.2
  width: 64 bits
  capabilities: fpu fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mm
x fxsr sse sse2 ss syscall nx pdpe1gb rdtscp x86-64 constant_tsc arch_perfmon nopl xtopology tsc_reliable nonstop_tsc cp
uid tsc_known_freq pni pclmulqdq ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt aes xsave avx f16c rdrand hypervi
sor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase tsc_adjust bmi1 avx2 smep bmi2 invpcid rdseed adx smap c
lflushopt xsaveopt xsavec xgetbv1 xsaves arat md_clear flush_l1d arch_capabilities
  configuration: microcode=4294967295
WARNING: output may be incomplete or inaccurate, you should run this program as super-user.
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Project 2: Implementation Workstation Solution

Part 3: Linux

Verify Network, Storage Information, and Memory

Network: ip addr show

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:70:05:0b brd ff:ff:ff:ff:ff:ff
    altnamename enp2s1
    inet 192.168.121.130/24 brd 192.168.121.255 scope global dynamic noprefixroute ens33
        valid_lft 1068sec preferred_lft 1068sec
    inet6 fe80::20c:29ff:fe70:58b/64 scope link
        valid_lft forever preferred_lft forever
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Storage Information: df -h

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            387M  1.9M  385M   1% /run
/dev/sda2        20G   9.4G   9.2G  51% /
tmpfs            1.9G   0     1.9G   0% /dev/shm
tmpfs            5.0M   8.0K   5.0M   1% /run/lock
tmpfs            387M  124K  387M   1% /run/user/1000
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Memory: -h

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi        1.1Gi        1.8Gi        36Mi        1.2Gi        2.7Gi
Swap:          3.6Gi          0B        3.6Gi
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Project 2: Implementation Workstation Solution

Part 3: Linux

Common CLI Operation

Verify the default root directory content

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ ls -l /
total 3819612
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Feb 26 2024 bin.usr-is-merged
drwxr-xr-x 3 root root 4096 Sep 26 15:36 boot
dr-xr-xr-x 2 root root 4096 Aug 6 02:17 cdrom
drwxr-xr-x 19 root root 4200 Sep 30 09:19 dev
drwxr-xr-x 138 root root 12288 Sep 26 15:40 etc
drwxr-xr-x 3 root root 4096 Sep 26 15:40 home
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Apr 22 2024 lib64 -> usr/lib64
drwxr-xr-x 2 root root 4096 Apr 8 2024 lib.usr-is-merged
drwx----- 2 root root 16384 Sep 26 15:29 lost+found
drwxr-xr-x 3 root root 4096 Sep 26 15:40 media
drwxr-xr-x 2 root root 4096 Aug 6 00:48 mnt
drwxr-xr-x 2 root root 4096 Aug 6 00:48 opt
dr-xr-xr-x 353 root root 0 Sep 30 09:18 proc
drwx----- 5 root root 4096 Sep 26 15:40 root
drwxr-xr-x 35 root root 880 Sep 30 09:20 run
lrwxrwxrwx 1 root root 8 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 Mar 31 2024 sbin.usr-is-merged
drwxr-xr-x 12 root root 4096 Aug 6 00:54 snap
drwxr-xr-x 2 root root 4096 Aug 6 00:48 srv
-rw----- 1 root root 3911188480 Sep 26 15:36 swap.img
dr-xr-xr-x 13 root root 0 Sep 30 09:18 sys
drwxrwxrwt 12 root root 4096 Sep 30 09:39 tmp
drwxr-xr-x 12 root root 4096 Aug 6 00:48 usr
drwxr-xr-x 14 root root 4096 Sep 26 15:39 var
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Create a file with the name "File1.txt" in root directory

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ sudo touch /File1.txt
[sudo] password for testvmubuntu:
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ ls -l /
total 3819612
lrwxrwxrwx 1 root root 7 Apr 22 2024 bin -> usr/bin
drwxr-xr-x 2 root root 4096 Feb 26 2024 bin.usr-is-merged
drwxr-xr-x 3 root root 4096 Sep 26 15:36 boot
dr-xr-xr-x 2 root root 4096 Aug 6 02:17 cdrom
drwxr-xr-x 19 root root 4200 Sep 30 09:19 dev
drwxr-xr-x 138 root root 12288 Sep 26 15:40 etc
-rw-r--r-- 1 root root 0 Sep 30 09:49 File1.txt
drwxr-xr-x 3 root root 4096 Sep 26 15:40 home
lrwxrwxrwx 1 root root 7 Apr 22 2024 lib -> usr/lib
lrwxrwxrwx 1 root root 9 Apr 22 2024 lib64 -> usr/lib64
drwxr-xr-x 2 root root 4096 Apr 8 2024 lib.usr-is-merged
drwx----- 2 root root 16384 Sep 26 15:29 lost+found
drwxr-xr-x 3 root root 4096 Sep 26 15:40 media
drwxr-xr-x 2 root root 4096 Aug 6 00:48 mnt
drwxr-xr-x 2 root root 4096 Aug 6 00:48 opt
dr-xr-xr-x 354 root root 0 Sep 30 09:18 proc
drwx----- 5 root root 4096 Sep 26 15:40 root
drwxr-xr-x 35 root root 880 Sep 30 09:20 run
lrwxrwxrwx 1 root root 8 Apr 22 2024 sbin -> usr/sbin
drwxr-xr-x 2 root root 4096 Mar 31 2024 sbin.usr-is-merged
drwxr-xr-x 12 root root 4096 Aug 6 00:54 snap
drwxr-xr-x 2 root root 4096 Aug 6 00:48 srv
-rw----- 1 root root 3911188480 Sep 26 15:36 swap.img
dr-xr-xr-x 13 root root 0 Sep 30 09:47 sys
drwxrwxrwt 17 root root 4096 Sep 30 09:39 tmp
drwxr-xr-x 12 root root 4096 Aug 6 00:48 usr
drwxr-xr-x 14 root root 4096 Sep 26 15:39 var
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Project 2: Implementation Workstation Solution

Part 3: Linux

Common CLI Operation

Under root directory, create a directory with the name Directory1

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ sudo mkdir /Directory1
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$ ls -l /
total 3819616
lrwxrwxrwx   1 root root      7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   2 root root    4096 Feb 26  2024 bin.usr-is-merged
drwxr-xr-x   3 root root    4096 Sep 26 15:36 boot
dr-xr-xr-x   2 root root    4096 Aug  6 02:17 cdrom
drwxr-xr-x  19 root root   4200 Sep 30 09:19 dev
drwxr-xr-x   2 root root    4096 Sep 30 09:51 Directory1
drwxr-xr-x 138 root root  12288 Sep 26 15:40 etc
-rw-r--r--   1 root root      0 Sep 30 09:49 File1.txt
drwxr-xr-x   3 root root    4096 Sep 26 15:40 home
lrwxrwxrwx   1 root root      7 Apr 22  2024 lib -> usr/lib
lrwxrwxrwx   1 root root      9 Apr 22  2024 lib64 -> usr/lib64
drwxr-xr-x   2 root root    4096 Apr  8  2024 lib.usr-is-merged
drwx-----  2 root root   16384 Sep 26 15:29 lost+found
drwxr-xr-x   3 root root    4096 Sep 26 15:40 media
drwxr-xr-x   2 root root    4096 Aug  6 00:48 mnt
drwxr-xr-x   2 root root    4096 Aug  6 00:48 opt
dr-xr-xr-x 354 root root      0 Sep 30 09:18 proc
drwx-----  5 root root    4096 Sep 26 15:40 root
drwxr-xr-x  35 root root    880 Sep 30 09:20 run
lrwxrwxrwx   1 root root      8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x   2 root root    4096 Mar 31  2024 sbin.usr-is-merged
drwxr-xr-x  12 root root    4096 Aug  6 00:54 snap
drwxr-xr-x   2 root root    4096 Aug  6 00:48 srv
-rw-----  1 root root 3911188480 Sep 26 15:36 swap.img
dr-xr-xr-x  13 root root      0 Sep 30 09:47 sys
drwxrwxrwt  17 root root    4096 Sep 30 09:39 tmp
drwxr-xr-x  12 root root    4096 Aug  6 00:48 usr
drwxr-xr-x  14 root root    4096 Sep 26 15:39 var
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:~/Desktop$
```

Move the File1 into the newly created directory

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/$ sudo mv File1.txt /Directory1
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/$ ls -l /Directory1
total 0
-rw-r--r-- 1 root root 0 Sep 30 17:06 File1.txt
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/$
```

Project 2: Implementation Workstation Solution

Part 3: Linux

Common CLI Operation

Insert your name into the file, save and exit once you are done

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ sudo nano File1.txt
[sudo] password for testvmubuntu:
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ cat File1.txt
Janson Gan
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$
```

Verify the content of the file

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ sudo nano File1.txt
[sudo] password for testvmubuntu:
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ cat File1.txt
Janson Gan
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$
```


Project 2: Implementation Workstation Solution

Part 3: Linux

Common CLI Operation

Verify the permission on the file

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ ls -ld File1.txt
-rw-r--r-- 1 root root 11 Sep 30 10:18 File1.txt
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$
```

Change the file permission to full access for owner, group and others

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ sudo chmod ugo+rw File1.txt
```

Verify the changed permission of the file

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ sudo chmod ugo+rw File1.txt
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$ stat -c "%a %A %n" File1.txt
777 -rwxrwxrwx File1.txt
testvmubuntu@testvmubuntu-VMware-Virtual-Platform:/Directory1$
```


Project 2: Implementation Workstation Solution

Part 3: Linux

Perform the command stat -c "%a %A %n"

```
testvmubuntu@testvmubuntu-VMware-Virtual-Platform: /Directory1$ stat -c "%a %A %n" File1.txt
777 -rwxrwxrwx File1.txt
testvmubuntu@testvmubuntu-VMware-Virtual-Platform: /Directory1$
```

Octal Value	Output
Stat	Linux command to display file status
-c	Set the specified format instead of the default; output a newline after each use of format.
%a	Access rights in octal format
%A	Access rights in human-readable form
%n	Access file name

Digit	User Type	Permissio n	Explanation
7	Owner	rwx	Read (4) + Write (3) + Execution (1)
7	Group	rwx	Read (4) + Write (3) + Execution (1)
7	Others	rwx	Read (4) + Write (3) + Execution (1)

Difference between touch, vi, vim, and nano command

touch	vi	vim	nano
Creates an empty file or updates the timestamp of an existing file.	Opens a file in the vi text editor (one of the oldest and most traditional editors in Unix/Linux).	□ An enhanced version of vi with more features.	Opens a file in the nano text editor (simple and user-friendly).

Best practices on directory and file permissions.

- Least privilege: Give only the access to people they need, no more.
- Use groups: Assign permissions to groups instead of individuals for easier management.
- Audit regularly: Check for files with overly open permissions and fix it.
- Backup configs: Keep a record of permission setting for recovery or troubleshooting.

Thank You